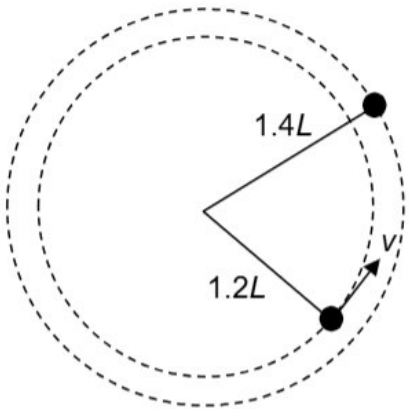
9 A mass is tied to an elastic string of unstretched length L which obeys Hooke's Law. The mass moves in a horizontal circle, about the other end of the string which is fixed, on a frictionless table. When its speed is v, the string extends to a length of $1.2L$.



What is the speed when the string extends to a length of $1.4L$?

A $0.86v$ **B** $1.2v$ **C** $1.3v$ **D** $1.5v$